What Ten Seeds Can and Cannot Resolve About Replay Strategy Selection Under Distribution Shift

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Abstract

Replay-strategy comparisons under non-stationarity — a design choice every continually adapting agent embeds — are normally reported as a ranking over policies, plus a claim that the ranking depends on the metric. We evaluate 8 adaptation policies at a fixed update and replay-sample budget across 5 drifting environments, with 10 seeds for each locomotion task and 5 for the process-control case. Selection-aware within-seed permutation tests separate best from worst in 3 of 5 environments after multiplicity correction, but the modal winner keeps rank one in as little as 0.47 of seed resamples. Testing all policy pairs before inspecting selected adjacencies protects the leaderboard analysis; at this seed count the exact-test p-value grid prevents any Holm rejection, so absence of rejection does not establish equal performance. A coarse recency-versus-non-recency contrast survives in 2 of 5 environments under a paired test with Benjamini–Hochberg correction. Pairing removes a shared seed effect carrying up to 0.91 of return variance. For metric disagreement, we estimate a concordant noise floor separately for each policy pair before averaging. 3 of 24 comparisons receive nominal BH flags, reduced to 2 of 12 when a related improvement metric is removed; these remain exploratory under model-dependent, nonsmooth inference. A Hopper-derived power simulation describes one uncorrected test and supplies no exclusion bound for other environments. Purely recent replay reverses its advantage across environments, and equal update/sample budgets leave wall-clock differing by 1.21–1.45$\times$ in this implementation.

1 Introduction

Off-policy agents deployed into changing environments have to decide what to replay. The literature offers uniform sampling, error-based prioritisation [Schaul et al., 2016], recency emphasis [Wang and Ross, 2019], and mixtures. A benchmark comparing them normally produces a ranking and an observation that it changes depending on whether one scores return, recovery, forgetting, or cost. We ask what a small fixed-budget study can support: 8 policies, 5 drifting environments, 10 seeds per locomotion task and 5 for the reactor.

The ordering needs selection accounting. Overlap of per-policy seed-bootstrap intervals is not a test of a difference. Tested within seed, with the post-hoc selection of the extremes built into the permutation, best and worst separate in 3 of 5 environments after Benjamini–Hochberg correction. Observed ranks also select the adjacent pairs. We therefore test all policy pairs and apply Holm before inspecting adjacencies. At this seed count the exact-test p-value grid precludes rejection (Section 3); this procedure cannot establish a full ordering, and its failure says nothing about equality.

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A coarse contrast survives, with a different testing target. Recency-biased replay beats non-recency replay in 4 of 5 environments by bootstrap interval, and in 2 (HalfCheetah, Hopper) under the exact paired test and BH correction. Between 0.38 and 0.91 of return variance is a seed effect common to every policy, which within-seed differencing removes. Group averaging also reduces residual noise. The adjacent-pair audit uses familywise control for selected comparisons, while the group contrast uses FDR control; their rejection counts do not compare like-for-like power.

Metric disagreement needs a pair-specific floor. A cross-metric flip rate needs a reference. Permuting a metric across policies gives chance, but two metrics that are both noise also flip at chance. We use each metric's split-half self-disagreement instead. For every policy pair we invert its self-disagreement rates and compose a concordant floor before averaging across pairs. 3 of 24 comparisons receive nominal Benjamini–Hochberg flags [Benjamini and Hochberg, 1995], reduced to 2 of 12 after removing a related metric. These are exploratory, model-dependent diagnostics; calibration checks the nonsmooth estimator rather than assuming validity (Section 5).

The power curve has a specific scope. In a Hopper-derived Gaussian model, detection is 1.00 for complete reversal and 0.27 at separation 0.21. This single uncorrected test illustrates sensitivity in that model. It supplies neither a uniform exclusion bound nor evidence of agreement for the table's unflagged cells (Section 6).

Two domain findings survive. Purely recent replay sits significantly above the other adaptive policies on HalfCheetah and below them on Ant, both directions surviving correction (Section 7); its rank-one seats elsewhere are descriptive. And the fixed-budget premise is definition dependent: updates and replay samples match exactly while wall-clock differs by 1.21–1.45$\times$ in this implementation, favouring the recency group where anything resolves.

Leaderboards, multi-metric protocols and fixed-budget comparisons are reporting conventions of continual- and lifelong-learning benchmarks. This small study separates observed contrasts from limits imposed by testing and estimation procedures.

2 Setup

Environments and drift. 5 environments: four MuJoCo locomotion tasks [Todorov et al., 2012] — Hopper, Walker2d, Ant, HalfCheetah — and a continuously stirred tank reactor from a process-control suite [Bloor et al., 2024], included because its drift is a loss of actuator authority rather than a change to locomotion mass or contact. The schedules differ by environment and escalate where they repeat: four friction events at 8k-step intervals on Walker2d, Ant and HalfCheetah (geom friction $\times 1.15$ rising to $\times 1.60$), one body-mass event on Hopper ($\times 1.25$), and two actuator-gain reductions on the reactor ($\times 0.85$, then $\times 0.70$). Every event falls in the first half of deployment, and every policy sees the same schedule within an environment.

Policies. 8 adaptation policies plus a Frozen control that performs no updates. Three bias sampling toward recent experience (RecentOnly, RecencyWeighted, StaticMix); five do not (Uniform, PER [Schaul et al., 2016], and three budgeted variants selecting by gradient norm, novelty, or TD error). All use the same underlying learner [Haarnoja et al., 2018].

Budget. Every adaptive policy receives an identical number of gradient updates and an identical number of replayed transitions; the audit confirms both are equal with zero variance across seeds and policies. This is the sense in which the comparison is same-budget, and Section 7 returns to what that phrase does and does not cover.

Metrics and seeds. Return, recovery AUC, forgetting, update cost, and improvement per thousand updates relative to Frozen. Forgetting here is the pre-drift reference return minus the lowest periodic evaluation during deployment, floored at zero, with lower better — a worst-case drawdown over the deployment run, not a re-evaluation on the pre-drift distribution. It is the weaker of the two senses the term carries, and Section 5 is about what this quantity does, not about retention of an earlier task. 10 seeds per policy per environment for the MuJoCo tasks, 5 for the process-control study, with the same seed set used for every policy — so the design is paired, which Section 4 shows is the decisive property.

Table 1: Best and worst adaptive policy per environment, interquartile mean of return with 95% seed-bootstrap intervals. The intervals describe per-policy seed variability; their overlap is not a test of a difference, so whether best and worst separate is tested in the text.

Environment	Seeds	Best	IQM [CI]	Worst	IQM [CI]
Hopper	10	RecentOnly	807.9 [667.2, 1008.3]	PER	641.2 [477.8, 801.1]
Walker2d	10	RecentOnly	507.3 [438.6, 612.5]	Uniform	417.5 [385.4, 466.2]
Ant	10	RecencyWeighted	649.9 [582.3, 747.0]	RecentOnly	517.3 [481.8, 554.2]
HalfCheetah	10	RecencyWeighted	3897.8 [3619.9, 4392.9]	PER	2949.4 [2782.8, 3227.4]
PC-Gym	5	RecencyWeighted	-695.5 [-1195.8, -360.0]	PER	-1044.3 [-1330.5, -496.0]

Training and reproducibility. All policies share the pretrained SAC agent and replay state within a seed. Actor and critics use 2 ReLU layers of width 256, batch size 256 and automatic entropy tuning. Appendix C gives learning rates, capacities, run lengths, evaluation and exact seed sets. Code and per-seed metrics are at `https://github.com/mandu5/rba-bench`. Reported numerical results and settings are exported from JSON artifacts into LaTeX macros; the source build regenerates analyses, calibration, figures and the manuscript without training or a simulator.

3 What the design can resolve

Not the ordering, for a repaired reason. Table 1 gives, per environment, the interquartile-mean return of the best and worst adaptive policy with 95% seed-bootstrap intervals. The intervals overlap in 3 of 5 cases — on Hopper the best policy is RecentOnly at 807.9 $[667.2, 1008.3]$ against PER at 641.2 $[477.8, 801.1]$ — which is where an analysis of this kind normally stops. That criterion is not a test of a difference, and it is biased against finding one: the seed effect that widens both intervals (Section 4) is shared, and cancels in a within-seed comparison that marginal intervals never make. Testing properly needs one more repair, because the extremes were selected on the data being tested. A paired sign-flip test on the selected pair rejects in 4 of 5 environments after Benjamini–Hochberg, but on Hopper a gap as large as the observed one arises in 0.24 of within-seed relabellings of the policies — the selection, not a difference. The label-permutation test whose statistic is the range of policy IQMs, so that the selection sits inside the permutation (Monte Carlo with the add-one correction, under the sharp null that policies are interchangeable within a seed), separates best from worst in 3 of 5 environments (Ant, HalfCheetah, PC-Gym).

The extremes separating is not the ordering resolving. Sorting the observed IQMs selects the adjacent pairs. We test all 28 policy pairs within each environment and apply Holm familywise correction [Holm, 1979] before inspecting the 35 selected steps; 0 reject. Crucially, with 10 seeds the smallest attainable two-sided p is 0.001953, above Holm's first cutoff 0.001786 at $\alpha = 0.05$. Rejection is impossible for this procedure even if every observed difference has the same sign. The reactor's smaller seed set is limited still further. Thus the formal count diagnoses the procedure's resolution, not equality of all policies or fundamental undetectability. Selected-pair-only BH gives 2 of 35 flags, reported as exploratory because it omits selection. The modal winner holds rank one in 0.47 (Walker2d) to 0.94 (HalfCheetah) of seed resamples, a descriptive measure of instability.

But one contrast, partly. Grouping the three recency-biased policies against the five others and differencing within seed gives an interval excluding zero in 4 of 5 environments, always in the same direction: 459.2 $[376.1, 546.1]$ on HalfCheetah with the sign consistent in 10 of 10 seeds, and 132.5 $[52.7, 201.6]$ on Hopper.

We do not get to stop there. Section 5 corrects a table of 24 tests for multiplicity, and this is a table of 5. An exact sign-flip permutation test on the per-seed differences, corrected with Benjamini–Hochberg, leaves 2 of 5 (HalfCheetah, Hopper); under Holm it leaves 1 (HalfCheetah). The bootstrap count of 4 is the uncorrected reading, and reporting it alone would be the practice this paper objects to.

The process-control study is a sharper case still. Its exact test returns $p = 0.1250$, but with 5 seeds the smallest *attainable* two-sided p is 0.0625: no effect size, however large, can reach 0.05. Its bootstrap interval excludes zero anyway, which is the interval being anticonservative rather than evidence the design could produce. Since 0.0625 is also the floor for any five-seed sign-flip comparison,

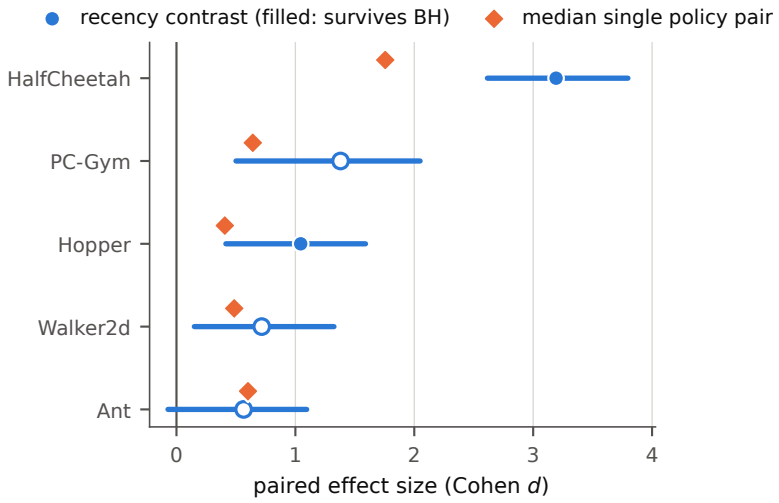


Figure 1: Why one claim resolves and the other does not, in common effect-size units. The paired recency contrast (circles, with 95% intervals) is larger than a typical single policy-versus-policy gap (diamonds) in every environment except Ant — the one environment where even the uncorrected contrast interval covers zero. Filled circles mark the contrasts that survive the exact test and multiplicity correction (HalfCheetah, Hopper); open circles clear at most the uncorrected bootstrap interval, and Ant's does not clear even that. Both quantities are paired within seed; the difference is that the contrast aggregates three policies against five, while a ranking is decided pair by pair.

this applies to the closest recent study [Korniak et al., 2026] as much as to us. The bound is specific to sign-flip tests, whose null has only 2^5 points; the label-permutation test of the extremes draws its null from within-seed relabellings of all 8 policies, which is how the same five seeds can still clear 0.05 there.

4 Why the contrast resolves and the ranking does not

The difference is not that one claim is luckier. Decomposing return into policy, seed and residual components puts the seed effect — the part that moves every policy together — at 0.38 (Ant) to 0.91 (PC-Gym) of total variance, the largest single component everywhere except Ant. A contrast differenced within seed removes it exactly; a ranking read off per-policy aggregates never differences at all. The selected adjacent steps are also paired, but their formal nonrejections are forced by the exact-test grid under all-pair Holm. Their small observed effect sizes are descriptive; that count alone cannot attribute failure to near-ties or establish the power of a different procedure.

Effect sizes make the same point (Figure 1). The paired contrast reaches $d = 1.04$ on Hopper against a median paired d of 0.41 for an individual policy-versus-policy gap, a factor of 2.57. The consistency check is Ant, the one environment where the ratio falls below one at 0.94 — and Ant is the one environment where even the uncorrected contrast interval covers zero, and the only one where the seed effect is not the dominant variance component — single-environment agreement, offered as consistency, not further evidence. The contrast has a second advantage beyond its size: it aggregates three policies against five, shrinking the residual noise, where an adjacent leaderboard step is a single near-tied pair with nothing to average over.

5 Metric disagreement needs a noise floor

Chance is an insufficient reference. Permuting one metric across policies gives chance agreement, but two metrics that are both sampling noise also cross-flip at chance. A raw flip count therefore does not by itself demonstrate a difference in latent orderings.

The pair-specific estimator. For policy pair j, let p_{Mj} be metric M's probability of misordering the pair on an independent half-sample, with $0 \leq p_{Mj} \leq 1/2$. Under independent half-sample errors its self-disagreement is $f_{Mj} = 2p_{Mj}(1 - p_{Mj})$. Two metrics sharing a latent ordering have concordant cross-disagreement $F_j = p_{Aj} + p_{Bj} - 2p_{Aj}p_{Bj}$. We estimate

$$\widehat{p}_{Mj} = \frac{1 - \sqrt{1 - 2\min(\widehat{f}_{Mj}, 1/2)}}{2}, \qquad \widehat{F} = \frac{1}{P}\sum_{j=1}^{P}(\widehat{p}_{Aj} + \widehat{p}_{Bj} - 2\widehat{p}_{Aj}\widehat{p}_{Bj}),$$

and report excess $\widehat{\Delta} = \widehat{C} - \widehat{F}$, where $\widehat{C}$ averages cross-flips of $A(h_1)$ versus $B(h_2)$ over disjoint halves and policy pairs. Each $\widehat{f}_{Mj}$ is first averaged over splits *for that pair*; inversion precedes averaging over pairs. Inverting a pooled self-rate gives a different estimator under heterogeneous pairwise noise. Averaging the two self-rates is different again, falling below F_j by $(p_{Aj} - p_{Bj})^2$ at the population level.

At 10 seeds we enumerate all 252 ordered balanced partitions of half-size 5. Delete-one replicates enumerate 630 ordered disjoint partitions of half-size 4, leaving one seed unused per partition. This removes split Monte Carlo error; it creates no new independent observations. The inversion assumes a continuous latent ordering, independent identically distributed seeds and independent half-sample errors; finite-sample self-rates may exceed one half and are clipped, not interpreted as proof of a chance ordering.

Inference and calibration. Seed uncertainty is summarised by a leave-one-seed-out jackknife with nominal Wald intervals and one-sided normal p-values. The statistic is nonsmooth, so these are model-dependent diagnostics, not automatically valid tests. Mistaking splits for independent evaluation units would instead create pseudoreplication [Hurlbert, 1984].

We calibrate the *same implementation* on 4 Gaussian models, each with 400 fresh 10-seed datasets: equal-noise concordance, policy-specific heterogeneous concordance, unequal heterogeneous concordance, and a heterogeneous partial reversal. Null type-I rates are 0.005, 0.030 and 0.050 at nominal $\alpha = 0.05$. Nominal 95% interval coverage is 0.978, 0.960, 0.960 and 0.930, respectively. The reversal's 0.930 coverage is below the nominal target; Monte Carlo intervals and model parameters appear in Appendix B. Neither these finite checks nor the largest observed null rate proves calibration for all benchmark environments.

The table covers the four locomotion environments; the 5-seed reactor is excluded because its two-seed halves give a particularly weak basis for the inversion. Update cost supplies no ranking under the exactly equalised budget. The remaining four metrics yield 24 cells. BH flags are retained as *nominal exploratory flags*; the calibration does not validate FDR control for arbitrary dependence among the actual metrics.

Related metrics and updated results. Within a seed, budget-normalised improvement is a positive affine transform of return, because both the Frozen reference and update count are shared. Their within-seed policy orderings match, but across-seed IQM trimming can change aggregate rankings. Removing this related metric leaves 12 comparisons; they are not asserted to be independent.

The revised estimator gives 5 nominal one-sided flags before multiplicity correction and 3 nominal BH flags: Hopper: return versus forgetting, excess +0.267 [+0.140, +0.395]; Hopper: forgetting versus budget-normalised improvement, excess +0.279 [+0.131, +0.428]; HalfCheetah: recovery versus forgetting, excess +0.090 [+0.029, +0.150]. Removing the improvement metric gives 2 flags among the 12 remaining comparisons: Hopper (return versus forgetting), HalfCheetah (recovery versus forgetting). All flagged cells involve drawdown-style forgetting, the quantity defined in Section 2; they identify candidate metric differences rather than establish a general need for multi-axis evaluation. Unflagged cells do not establish agreement. In 24 of 24 cells at least one policy pair is clipped (336 of 672 pair-by-cell estimates), an additional finite-sample limitation rather than saturation of every cell's mean floor.

6 What the power diagnostic covers

We simulate independent Gaussian metric observations with the latent spacing of Hopper returns and a common noise SD of 140.58, derived from the observed SDs of within-seed policy differences.

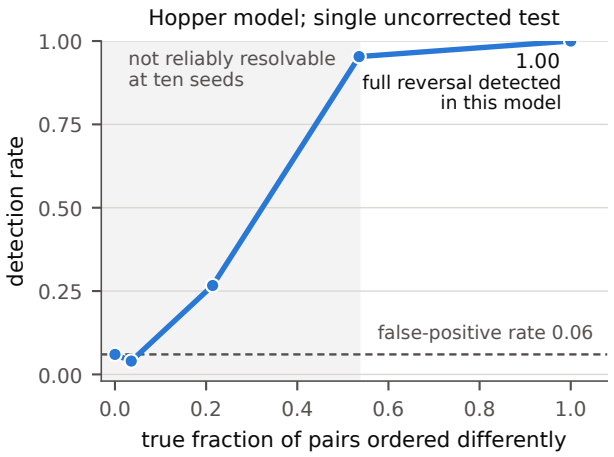


Figure 2: Detection in a Hopper-derived Gaussian model at 10 seeds for a single uncorrected test. The zero-separation point estimates false-positive rate on draws independent of threshold calibration. The curve supplies no exclusion bound for the other environments or for the multiplicity-corrected metric table.

Reversing a growing block of latent policy values controls the true fraction of discordant pairs. The statistic uses the same pair-specific floor and exhaustive splits. A threshold is the upper null quantile of 300 calibration draws; each plotted point uses 300 *independent* evaluation draws.

Detection is 0.06 at zero separation, 0.27 at separation 0.21, and 1.00 under full reversal. The first sampled gridpoint with detection at least 0.8 is 0.54; we do not infer an exact threshold between gridpoints. The illustrative model is homoscedastic and covers one environment and one uncorrected procedure. The metric table uses heterogeneous quantities, estimated floors and BH flags, so this curve does not establish their effective power. In particular, a non-significant cell does *not* uniformly exclude gross disagreement across the four environments. Environment- and metric-specific power or valid inversion of tests would be required for such a bound. No agreement or equivalence conclusion is made.

7 What survives

Purely recent replay reverses across environments. A rank is not a result, so we test the claim directly: the per-seed difference between RecentOnly and the mean of the other adaptive policies, with the same exact sign-flip test and Benjamini–Hochberg correction used elsewhere. It is above the others on HalfCheetah (347.6, $d = 2.16$, positive on 10 of 10 seeds, $p = 0.0020$) and below them on Ant (-86.5, $d = -1.28$, positive on 0 of 10 seeds, $p = 0.0020$). Both survive correction across 5 environments, so the reversal is resolvable in both directions — the only claim here that resolves with opposite signs in different environments. Its rank-1 position on Hopper is descriptive only ($p = 0.0547$), as is Walker2d's ($p = 0.0820$).

Recency without an anchor is not a strategy that degrades gracefully; it is one that inverts. We report the phenomenon rather than proposing a method to fix it, because at this seed count we could not distinguish a proposed fix from the policies already present.

"Same budget" is definition dependent. Update counts and replay samples are identical across policies with zero variance. Wall-clock is not: it spreads by 1.21–1.45×, grouped the same way in every environment — RecentOnly and StaticMix cheapest, PER and the budgeted variants dearest — though the full ordering permutes within the groups. The quantity is the summed time of batch selection plus one learner update, over every deployment update, in a single CPU process; the grouping tracks how a policy draws its batch: RecentOnly and StaticMix index into the buffer, while the other six — Uniform included — score every stored transition before drawing. A different implementation or accelerator would move the ratio, so we read only its sign and its grouping, not its magnitude. The direction matters at the level the surviving claim lives at: the recency group is

cheaper than the non-recency group in every environment (1.09 against 1.30 on Hopper, and likewise elsewhere), so a wall-clock-matched budget would widen the recency advantage rather than explain it away. Korniak et al. [2026] report the latency profile of PER directly; the point here is narrower and about accounting — a benchmark that calls itself fixed-budget has fixed one resource and left another free, and which one it picked is not neutral.

8 Related work

Korniak et al. [2026] is the closest study. It decomposes replay into volume, expected recency and sampling entropy, and finds non-uniform replay most useful when replay volume is low. Two differences matter. It studies stationary tasks — distribution shift appears in its future work, not its experiments — whereas every environment here drifts mid-deployment. And it operates at a much lower replay volume than we do, a regime in which it predicts recency bias to be approximately neutral; we observe a recency advantage anyway, which suggests the mechanism under drift is staleness rather than sample efficiency. We do not claim "recency beats uniform" as a finding: that is theirs, at larger scale.

Benchmarks for continual and lifelong RL, such as Wołczyk et al. [2021], standardised the task suites and the forgetting and transfer metrics this literature reports. Our question is downstream of that: given such a suite, which of the comparisons it invites are supported by the number of seeds people run on it.

On statistical practice, Agarwal et al. [2021] established interval reporting and robust aggregates as the norm in RL evaluation, and Henderson et al. [2018] documented how far conclusions move with seeds. Our addition is narrower: for the specific statistic of cross-metric ranking disagreement, the reference distribution is neither zero nor chance but the metric's own split-half noise, and once that is supplied the statistic turns out to be underpowered at the sample sizes in use.

9 Limitations

This study covers four locomotion tasks at 10 seeds and one reactor at 5; the split-half metric analysis covers only the former. The halves are subsets of existing runs, not additional independent training replicates. Each environment uses one shift family, and repeated events escalate monotonically, with no return to earlier dynamics. Nothing here establishes behaviour under recurring or cyclic shifts, different learners, or foundation-model agents. Forgetting denotes deployment drawdown rather than retention on an earlier task distribution.

Exact sign-flip inference requires independent seeds and a null invariant to signs of the paired differences; label permutation requires the sharp within-seed interchangeability null. All-pair Holm accounts for adjacent selection but is unable to reject at this seed count. The formal 0/35 count therefore measures this procedure's p-value resolution, not evidence that every policy comparison is equal. The coarse contrast is a group comparison and does not identify a preferred recency policy. Its FDR target differs from the selected-pair audit's familywise target.

The pair-specific floor assumes independent half-sample errors around continuous latent orderings. Finite-sample inversion, clipping and the nonsmooth jackknife can introduce bias or undercoverage. The 4 simulation cases give 0.930 coverage in the partial-reversal model at a nominal 95%, and their 400 datasets each leave material Monte Carlo uncertainty. Independent Gaussian policy/metric errors do not exhaust the benchmark's possible dependence or heterogeneity. Consequently, the nominal Wald/BH flags remain exploratory. The separate Hopper power curve cannot exclude large differences for all other metrics or environments.

The three cross-environment test families share data; joint BH across their 15 tests leaves the reported survivors unchanged. This sensitivity does not validate arbitrary post-hoc hypotheses. Wall-clock comparisons describe this implementation and hardware mix rather than a compute-optimal policy.

10 Conclusion

Under identical adaptation update and replay-sample budgets, the paired recency-group contrast survives BH in 2 of 5 environments, and purely recent replay reverses direction across tasks. Selection-aware tests distinguish best from worst in 3 of 5 environments, but a complete leaderboard is not established: all-pair Holm is resolution-limited at the observed seed counts. Metric dependence also needs an explicit estimator and noise model. Separately inverting each policy pair's split-half self-disagreement gives 3 nominal exploratory flags among 24 metric comparisons. Calibration exposes the estimator's finite-sample limits; a single Hopper power curve supplies no benchmark-wide exclusion bound. These checks use existing per-seed results, and make the scope of the evidence explicit without inferring equality from nonrejection.

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A All environment-by-metric-pair comparisons

Table 2 gives every cell behind Section 5. *cross* is the flip rate between one metric on one seed half and the other metric on the other half; *floor* is the concordant rate implied by the two metrics' separately inverted policy-pair self-disagreement over the same halves (Section 5). Intervals are leave-one-seed-out jackknife Wald intervals with nominal 95% coverage. A star marks a nominal exploratory Benjamini–Hochberg flag at $\alpha = 0.05$; neither distribution-free interval validity nor table-wide FDR control is asserted. Unstarred rows do not establish agreement.

Table 2: Cross-metric disagreement against the mean pair-specific concordant floor. Intervals and stars are nominal exploratory diagnostics.

Environment	Metric pair	cross	floor	excess	nominal 95% CI	
ant	forgetting vs. improvement/1k	0.333	0.383	-0.051	[-0.167, +0.066]	
ant	recovery vs. forgetting	0.377	0.380	-0.004	[-0.069, +0.061]	
ant	recovery vs. improvement/1k	0.238	0.250	-0.012	[-0.055, +0.031]	
ant	return vs. forgetting	0.336	0.393	-0.057	[-0.159, +0.045]	
ant	return vs. improvement/1k	0.216	0.200	+0.016	[-0.019, +0.051]	
ant	return vs. recovery	0.275	0.274	+0.002	[-0.028, +0.031]	
halfcheetah	forgetting vs. improvement/1k	0.253	0.303	-0.050	[-0.153, +0.053]	
halfcheetah	recovery vs. forgetting	0.410	0.320	+0.090	[+0.029, +0.150]	⋆
halfcheetah	recovery vs. improvement/1k	0.222	0.259	-0.037	[-0.098, +0.024]	
halfcheetah	return vs. forgetting	0.247	0.301	-0.054	[-0.165, +0.058]	
halfcheetah	return vs. improvement/1k	0.047	0.045	+0.002	[-0.021, +0.025]	
halfcheetah	return vs. recovery	0.218	0.253	-0.035	[-0.107, +0.036]	
hopper	forgetting vs. improvement/1k	0.645	0.366	+0.279	[+0.131, +0.428]	⋆
hopper	recovery vs. forgetting	0.427	0.446	-0.019	[-0.079, +0.041]	
hopper	recovery vs. improvement/1k	0.541	0.454	+0.086	[-0.184, +0.357]	
hopper	return vs. forgetting	0.635	0.368	+0.267	[+0.140, +0.395]	⋆
hopper	return vs. improvement/1k	0.325	0.288	+0.037	[-0.036, +0.111]	
hopper	return vs. recovery	0.534	0.462	+0.072	[-0.225, +0.368]	
walker2d	forgetting vs. improvement/1k	0.614	0.402	+0.212	[-0.055, +0.479]	
walker2d	recovery vs. forgetting	0.423	0.406	+0.018	[-0.027, +0.062]	
walker2d	recovery vs. improvement/1k	0.535	0.383	+0.152	[-0.031, +0.335]	
walker2d	return vs. forgetting	0.627	0.377	+0.250	[+0.007, +0.492]	
walker2d	return vs. improvement/1k	0.329	0.297	+0.032	[-0.037, +0.101]	
walker2d	return vs. recovery	0.531	0.371	+0.160	[-0.005, +0.325]	

B Calibration design and Monte Carlo uncertainty

Calibration and benchmark analysis share the implementation of the pair-specific estimator. Each case draws 400 independent datasets of 8 policies and 10 seeds. Latent policy means are evenly spaced from one to zero. Equal noise uses SD 0.35 for both metrics. Heterogeneous noise uses 0.35 times a geometric sequence from 0.25 to 2.5 across policies, reversed for the other metric. The unequal case additionally multiplies the second metric's SDs by 3.4. The partial reversal reverses the first half of the latent policy means. Errors are independent across seeds, policies and metrics.

For concordant nulls the population excess is analytically zero. Coverage is checked against zero, not against the sampling mean of a biased estimator. For the reversal, 100,000 separately seeded population half-samples approximate the target as 0.0879 (Monte Carlo SE 0.0003). Its interval coverage is checked against that target; the small target approximation error is not included in the coverage interval. Table 3's brackets are Wilson 95% Monte Carlo intervals for proportions. Rejection is one-sided at $\alpha = 0.05$: type-I error for the concordant rows, power for the reversal. The cases are fixed model checks, not tuned to produce a pass.

Table 3: Finite-sample calibration of the revised estimator. Nominal interval coverage is 95%; rejection rate alone is not a calibration guarantee.

Model	Bias	Coverage [MC CI]	Rejection [MC CI]
Equal noise, concordant	-0.010	0.978 [0.958, 0.988]	0.005 [0.001, 0.018]
Heterogeneous, concordant	-0.000	0.960 [0.936, 0.975]	0.030 [0.017, 0.052]
Heterogeneous, unequal	+0.028	0.960 [0.936, 0.975]	0.050 [0.033, 0.076]
Heterogeneous, reversal	-0.002	0.930 [0.901, 0.951]	0.390 [0.343, 0.439]

C Training setup and reproducibility scope

The actor and both critics are MLPs with 2 hidden ReLU layers of width 256, a squashed Gaussian actor and Adam learning rates 0.0003 (actor) and 0.0003 (critic). SAC uses discount 0.99, target mixing 0.005, automatic entropy tuning with target negative action dimension, batch size 256, actor updates when the global step is divisible by 2, and target updates every 1 global step. Random-action warmup lasts 5,000 environment steps. Pretraining performs an update per step once replay contains that many transitions; policy variants then share its checkpoint and replay state within each seed. Deployment uses the windowed update budget in Table 4. Frozen receives no deployment updates.

Evaluation is deterministic with 3 episodes per scheduled evaluation and a seed offset of 10,000; evaluation intervals are listed in the table. Seed sets are 21–30 (Hopper), 31–40 (Walker2d), 51–60 (Ant), 41–50 (HalfCheetah), and 61–65 (reactor). Every policy in an environment uses the same set. The observed per-run deployment updates are 35,000 on Hopper, 45,000 on the other locomotion tasks, and 20,000 on the reactor; replay samples equal updates times batch size.

Table 4: Declared configurations, checked against observed seed grids and update/sample budgets. Buffer/recent are transition capacities; window/updates gives environment steps and the allowed updates per window.

Environment	Seeds	Pretrain/deploy	Buffer/recent	Window/updates	Eval interval
Hopper	10	50,000/70,000	200,000/30,000	2,000/1,000	2,000
Walker2d	10	60,000/90,000	250,000/35,000	2,500/1,250	2,500
Ant	10	60,000/90,000	250,000/35,000	2,500/1,250	2,500
HalfCheetah	10	60,000/90,000	250,000/35,000	2,500/1,250	2,500
PC-Gym	5	40,000/40,000	200,000/30,000	1,000/500	2,000

The MuJoCo source configs are `configs/train/*_v21_main.yaml`; the reactor uses `configs/train/pcgym_case_study.yaml`, including its normalisation, integrator, reward and constraint parameters. The small reactor case is not evidence of general process-control behaviour. The released per-seed CSVs suffice to reproduce analyses and manuscript artifacts without training. Training itself requires the task simulators and additional compute; the analysis rebuild does not establish bitwise retraining reproducibility.